

A Practical Sanity Check of GiBleed

Recovering an undocumented disease module from a synthetic-data generator's output — using only SQL and clinical reasoning.

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Summary

Using seven SQL queries and clinical reasoning, I audited GiBleed — the Synthea-generated (Walonoski et al., 2018) Eunomia dataset (OHDSI/Eunomia, 2026) on which most people first learn the OMOP Common Data Model — and recovered an **undocumented generator module from its output alone**, without reading Synthea's source or documentation. Eight conditions occur exactly once per person and never again; seven of them, read together, are the **differential diagnosis of gastrointestinal (GI) bleeding**, and the eighth is osteoarthritis, present in 100% of the population. All eight are confined to a ~15-year midlife window and stop by age 47, in a database that records other conditions to age 109 and in which **no one dies**.

None of this violates a range check. Every value is well-formed, in-range, type-correct, and conformant — the dataset passes the Kahn et al. conformance/completeness/plausibility framework (Kahn et al., 2016) and the OHDSI Data Quality Dashboard. The artifacts are invisible to the structural tooling and visible only from a combined clinical-biomathematical vantage point.

The thesis in one line. The mathematics tells me what topological shape a disease should leave in data; the medicine tells me whether the shape aligns to it or not.

This is a self-directed learning artifact, not a publication. Its purpose is to help familiarize myself with the OMOP-CDM ecosystem and also to demonstrate, on the ecosystem's own canonical teaching dataset, the multidisciplinary sanity check that structure-only tooling cannot perform — and that a federated research network needs performed **at each source**.

1 · Why I did this

I wanted to be genuinely comfortable inside the OMOP/OHDSI ecosystem — not merely familiar with it. The gap was practical rather than conceptual as I knew the epidemiology and the modelling aspects, and I needed fluency with the stack. So I did what I would do with any

unfamiliar instrument — I ran a learning-by-diong session, through performing an actual sanity check to gain hands-on experience.

2 · The observation that started it

Within an hour of loading the Eunomia GiBleed dataset (OHDSI/Eunomia, 2026) — synthetic, realistic, and the default sandbox where most people first meet OMOP — one thing stopped me:

The dataset has 2,694 patients. It also has exactly 2,694 cases of osteoarthritis.

Osteoarthritis is a chronic, degenerative disorder of articular cartilage driven by mechanical wear over time; its principal risk factors are age, obesity, female sex, and prior joint trauma. At the population level, symptomatic osteoarthritis most frequently occurs after age 40, and prevalence rises steeply with age (Steinmetz et al., 2023). A 100% prevalence figure is therefore not a data point but a claim about a population — that *everyone here is old enough to have worn out a joint*.

That produced a prediction, before I ran a single further query:

If every person has a degenerative disease, this population should contain almost no paediatric patients.

The prediction held, and the mechanism was stranger than expected. Every osteoarthritis record in the database is written at age ≥ 31 (Table 2; Figure 3). Since all 2,694 people have one, all 2,694 people reached 31 year of age. There are no children in GiBleed — only childhood *records* belonging to adults who later grew up. That anomaly was the whole beginning, and it told me a multidomain sanity check of this teaching dataset would let me learn the CDM by doing while producing something useful.

3 · The lens I brought

My assumption at the outset was that GiBleed would be structurally and numerically flawless which turned out to be a correct assumption. However, structural validity is one question; a representative dataset must also satisfy properties that automated conformance checks never test — chiefly **representativeness** of the source population, **absence of systematic bias**, and **independence** of observations. Establishing those requires three overlapping disciplines, with *fitness for purpose* at the centre.

Table 1. The three-domain sanity-check framework.

Domain	The question it asks	GiBleed
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Domain	The question it asks	GiBleed
Computer science	Is the value well-formed?	✓ Passes (conformant)
Medicine	Is this what the disease pathophysiology naturally does?	✗ (this audit)
Biomathematics	Is this a topological shape a disease can leave?	✗ (this audit)

Automated data-quality tooling clears only the first circle. Every finding below sits in the **Medicine** $\cap$ **Biomathematics** overlap, where no range check looks. The centre — fitness for purpose — requires all three, and the OMOP CDM has no field for it. That intersection is what I bring to a research team, and it is the argument of this repository.

4 · The two questions

- **From medicine:** does this dataset behave like the disease process — in real-world prevalence, risk stratification, natural history, pathophysiology, prescribing, complications, and differential diagnosis?
- **From biomathematics:** treating each condition as a compartment in a deterministic system, what is the *topological shape* of the disease states and the *flow dynamics* between them? A condition's transition structure leaves a signature in the record counts. The instrument for this second question is a column most people treat as bookkeeping: **records per person**.

Everything below comes from seven SQL queries against the 2,694-person Eunomia database (OHDSI/Eunomia, 2026) (`sql/audit-queries.sql`). No machine learning or privileged access and can be executed in roughly ninety seconds.

5 · Findings

5A · The medical lens

Most consequential first. Each finding carries the downstream analytic implications it invites by departing from the population should represent.

M1 · There is no confounding by indication

355 of 479 GI-bleed patients are on celecoxib (74.1%), and in every case the drug precedes the bleed — 355 times out of 355, with no reversals and no same-day ties.

In the real world, patients prescribed celecoxib are precisely those at elevated GI risk: because a COX-2-selective agent spares the COX-1-dependent gastric mucosa, it is preferentially channelled to patients with prior gastropathy, gastroprotective-agent use, and

advancing age. Coxib users accordingly carry a greater underlying disease burden than users of nonselective NSAIDs. This is the definition of **confounding by indication** (Kyriacou and Lewis, 2016) — the reason NSAID→GI-bleed is a hard causal question rather than a division problem.

Downstream implications — the most consequential finding in this audit. Every standard confounding-control method runs cleanly on GiBleed: propensity scores and inverse probability of treatment weighting (IPTW) (Austin, 2011), new-user designs (Ray, 2003), negative controls (Lipsitch et al., 2010), and self-controlled designs (Whitaker et al., 2006) all execute, converge, and return plausible numbers. Each of these methods exists to *subtract a bias*. GiBleed contains none, so each one subtracts zero and returns the number it was handed. Below are three consequences that would follow:

- **RR = 1.32 is not a measurement.** With no confounding and no noise, the risk ratio (OR = 1.40) reflects the generator's parameter settings — not biology, not bias, not celecoxib.
- **Benchmarking here is uninformative.** A new causal-inference method tested against GiBleed will appear to work; that conclusion is unsupported, because it shows the code executes, not that the method removes bias.
- **The diagnostics are untested too.** There are no positivity violations, no extreme weights, and no overlap failures — the pathologies these checks exist to catch never occur.

M2 · 100% osteoarthritis prevalence

Every person carries osteoarthritis, with exactly one record each (2,694 of 2,694). Real-world symptomatic osteoarthritis prevalence is roughly an order of magnitude lower — on the order of ~13% among adults aged 55 and over (Steinmetz et al., 2023).

Downstream Implications. No unexposed comparator exists — there is no osteoarthritis-free group to compare against. With zero variance, osteoarthritis has zero covariance with every other variable; it cannot confound, mediate, stratify, or be adjusted for. **Formally, it is not a variable but a constant** — indeed, the sample space itself. Any model entering it as a covariate is rank-deficient in that dimension (the column is identical to the intercept), so its coefficient is not separable from the baseline. Every comorbidity overlap involving osteoarthritis is therefore a forced requirement: for example, Otitis media $\cap$ Osteoarthritis = 2,025, which is *all* 2,025 otitis patients. The co-occurrence carries no added information since it is arithmetically guaranteed and serves only to confirm the 100% figure and the integrity of the person-level joins.

M3 · The GI-bleed differential is present, complete, and one-shot

Eight conditions occur exactly once per person and never again. Read as a clinician, seven of them are the differential diagnoses of GI bleeding.

Table 2. The eight one-shot conditions of the GI-bleed module.

GI section	Condition	People (n)	Prevalence	Age window	Role
Upper GI	Peptic ulcer	802	29.8%	24–34	Cause
Upper GI	Esophagitis	409	15.2%	31–44	Cause
Upper GI	Angiodysplasia of stomach	388	14.4%	30–44	Cause
Lower GI	Diverticular disease	405	15.0%	32–47	Cause
Lower GI	Polyp of colon	380	14.1%	31–46	Cause
Lower GI	Ulcerative colitis	413	15.3%	31–45	Cause
Outcome	Gastrointestinal hemorrhage	479	17.8%	32–47	Outcome
Indication	Osteoarthritis	2,694	100%	31–47	Indication

This is the list one would generate at the bedside and use during their patient work-up to rule-out differentials and arrive at a working or definitive diagnosis. They are instantiated as a family of one-shot events.

Downstream Implications. Nobody bleeds twice in the dataset. Rebleeding is *the* clinical problem in GI hemorrhage, and it does not exist here, so time-to-recurrence, rebleed risk, and secondary prevention are unstudyable. Each cause appears once, in isolation, with no severity gradient and no progression, so etiology-specific risk cannot be modelled: ulcerative colitis with no flares, diverticular disease with no diverticulitis — despite ~20% of incident diverticulitis recurring in reality (Strate and Morris, 2019) — and peptic ulcer with no *H. pylori* eradication or relapses.

A note on two columns that are easily conflated. `recs_per_person = 1.00` is not 100% prevalence. Ulcerative colitis affects 15.3% of people, and all 413 of them got it exactly once. Only osteoarthritis is at 100% prevalence.

M4 · The age windows are clinically improbable

Every module member is diagnosed inside a ~15-year midlife window and never after 47 years — in the database that records other conditions out to 109.

- **Angiodysplasia of the stomach** is the most common vascular abnormality of the GI tract and the second leading cause (after diverticulosis) of lower-GI bleeding in patients over 60 years, typically in the context of aortic stenosis and acquired von Willebrand syndrome (Heyde syndrome) (Hudzik et al., 2016). Here it appears in 388 people aged 30–44 years old, and not again.
- **Diverticular disease** rises with age from roughly 5% at 40 to about 65% by 85 (Strate and Morris, 2019). Here it stops at 47 years of age.

- **Polyp of colon** — colorectal screening now begins at 45 years precisely because incidence climbs through and beyond that age (US Preventive Services Task Force, 2021). In the database polyps are finished by 46 years old.
- **Osteoarthritis** rises steeply after 55 and does not switch off (Steinmetz et al., 2023). Here there are now new cases recorded after 47 years of age.

This is unlikely to be censoring, and the proof is in the same table: viral sinusitis is recorded from age 0 to 109 years (Figure 3). The recording apparatus demonstrably works across the whole lifespan, so the module's silence after 47 years of age is a true absence.

Downstream Implications. Age cannot confound within the module, so age-adjusted models are unidentifiable; age-stratified analysis is meaningless because there are no strata above 47 years; incidence-by-age curves are step functions with an 'off switch', which no real degenerative or vascular process produces. Any study relying on age as an effect modifier will find no effect — and the null will be an artifact.

A precision point I want to be exact about. The pathology is *not* an inverted age gradient. Prevalence rises from 0% to $k\%$ across the window and then plateaus — monotone increasing, exactly as a chronic, irreversible condition should behave. What is wrong is that **incidence** has an off switch and the window is shifted young and truncated at 47 years; **prevalence** does not fall, because no one recovers and no one dies.

M5 · No deaths are recorded

There are zero deaths, yet 98 people are over 100 years old (3.6%, against a real-world figure of ~0.03% (Schaeffer, 2024)), and the population reaches 110 years. A population spanning every age to 110 years with no death record is implausible.

Downstream Implications. Competing risk does not exist — in real GI-bleed research, death competes with rebleeding, so cumulative incidence and naive risk diverge; here they coincide. Survival analysis is vacuous because there are no events. Immortal time bias is undetectable because everyone genuinely is 'immortal' (Suissa, 2008). No mortality-adjusted or death-censored design can be tested against the data.

M6 · Prescribing patterns differ from the usual prescribing norms

The dataset shows 68% celecoxib exposure and 98.7% tetanus/diphtheria toxoid coverage. Celecoxib is a second-line, cardiovascular-risk-carrying, selectively channelled COX-2 inhibitor; at the population level, coxibs are a small, selected share of NSAID prescribing (Roumie et al., 2005). A 68% population exposure is therefore implausible by roughly an order of magnitude. Likewise, only about 59% of U.S. adults report a tetanus-toxoid-containing vaccination in the past decade (CDC, 2025), against 98.7% here.

Downstream Implications. No one was switched to a COX-2-selective agent *because* they were high-risk, so channelling — the real-world driver of confounding by indication — is

absent. One exposure record per person compresses years of chronic NSAID therapy, the mechanism by which GI risk accumulates, into a single event. And near-universal adherence makes adherence research and healthy-adherer bias unstudyable.

M7 · There is no comorbidity structure

Every pairwise overlap equals the smaller group entirely — a consequence of osteoarthritis behaving as a constant rather than a variable (M2).

Downstream Implications. Charlson and Elixhauser comorbidity indices are effectively constants. Multimorbidity research is impractical, and covariate adjustment is vacuous because there is nothing to adjust for. In reality, a 70-year-old with osteoarthritis typically also carries hypertension, diabetes, or chronic kidney disease in many cases; here the conditions do not interlock.

5B · The biomathematical lens

Each condition is treated as a compartment in a deterministic system, which implies transition dynamics between states.

B1 · Eight absorbing states that should not exist

recs_per_person is not bookkeeping; it is a measurement of the disease's state graph. A value of 1.00 signifies an **absorbing state** — entered once, never re-entered, never left. For a chronic, recurrent, or progressive condition this is an improbable topological shape: the transition matrix has a single entry and no self-loop. The evidence is the contrast — most conditions recur, tracing the transitions a real disease process leaves behind.

Table 3. Records per person across representative conditions.

Condition	Records per person
Viral sinusitis	6.43
Acute viral pharyngitis	3.92
Acute bronchitis	3.22
Otitis media	1.78
Sprain of ankle	1.41
Chronic sinusitis	1.02
The eight module members	1.00

Twenty-three conditions behave like real life; eight are pinned flat at 1.00 (Figure 1). One number alone proves nothing — the neighbours make it legible. Viral sinusitis at 6.43 is the positive control: the apparatus can represent recurrence, so the flatness of the eight is a design decision and not an instrument limit.

Downstream limitations. Multi-state and Markov models are unidentifiable, because no transition rates can be estimated when no transitions occur; there are no rate parameters, sojourn times, or hazards of re-entry. Hierarchical and mixed models have no within-person variance — with one observation per person, the random-effect variance component cannot be separated from the residual, so random effects are unidentifiable. And because the join cardinality is 1:1, fan-out also does not occur.

B2 · The death compartment was never built

Death is the canonical absorbing state of any human-population model — every $S \rightarrow I \rightarrow R$ system, every Markov chain over a lifespan, every survival model has one. GiBleed's is missing entirely while the population ages to 110 years.

Downstream limitations. This is an open system with no exit; population is not conserved, so no balance- or carrying-capacity model can be fitted, and any compartmental model with a mortality sink is unidentifiable. The absence is invisible to a conformance check, because an empty table is still a valid table.

B3 · The hazard function has an off switch

For osteoarthritis, incidence is zero before ~31 years of age, positive inside the window, and zero after 47 years — in a system that demonstrably records events to 109 year.

Downstream limitations. No parametric survival family can fit this. The Weibull, Gompertz, and exponential families all assume a hazard that is monotone or at least continuous; a hazard that terminates is a rectangular pulse. Real degenerative and vascular hazards accumulate with age, whereas this one is switched off abruptly. As noted in M4, the gradient is *not* inverted — prevalence rises and then plateaus, monotone and correct; the defect is that incidence has an off switch and the window is shifted young and truncated. A fitted survival model will nonetheless return parameters with standard errors — the best possible summary of a shape the family cannot represent — which is precisely the kind of silent, plausible-looking failure that this sanity check would identify and reveal.

B4 · Determinism where stochasticity belongs

Exposure precedes outcome 355 times out of 355. The `>=` in the ordering test folds same-day cases into `bleed_first`, so zero means no reversals *and* no same-day ties. Real data usually carries noise, misclassification, backfilled coding, reverse-time artifacts, and same-day collisions; real ordering runs ~95–98% with a ragged tail.

Downstream limitations. There is no measurement error to model and no misclassification to correct. Window-based cohort definitions — index dates, lookback periods, tie-breaking rules — are never stressed, and any method robust to temporal noise is untested here.

B5 · Two independent signatures with one boundary and no overlap

recs_per_person and the age envelope were measured separately, yet they select the same eight conditions.

Table 4. Two independent signatures selecting the same eight conditions.

Signature	The eight module members	The other twenty-three
Records per person	1.00	1.02 – 6.43
Age window (max – min)	~15 years (none after 47)	up to 96–110

Nothing sits between them except one partially-scripted case — *E. coli* UTI (1.15 records/person, ages 16–66). Two unrelated rulers drawing the identical line do not describe a disease; they likely describe a code path (Figure 2).

5C · Reverse-engineering Synthea's internal thresholding

I never read Synthea's source code, module definitions, or documentation. Everything below was recovered from output alone — from 65,332 condition records and seven queries.

Table 5. Generator internals recovered from output alone.

Recovered	Evidence
A discrete GI-bleed module of 8 conditions exists	Two independent signatures select the same 8, with no overlap
Its membership is the clinical differential of GI hemorrhage (3 upper, 3 lower, plus the outcome)	The identity of the eight
One-shot semantics — no recurrence loop	recs_per_person = 1.00, exactly, eight times
An onset age gate at ~30–32	min ages: 30, 31, 31, 31, 31, 32, 32
A termination gate at ~44–47	max ages: 44, 44, 45, 46, 47, 47, 47
Peptic ulcer runs a distinct, younger gate (24–34)	~7 years below the rest of the module
Osteoarthritis is the eligibility scaffold, not a modelled disease	100% prevalence + identical topology + identical window
No mortality module is wired into this scenario	death empty; ages to 110
Module boundaries are hard, not probabilistic	<i>E. coli</i> UTI at 1.15 is the only intermediate case in 31 conditions

I then checked the metadata. The `cdm_source` table independently confirms the generator: Synthea (Walonoski et al., 2018), OMOP CDM v5.3.1, ETL-Synthea (OHDSI/ETL-Synthea, 2026), released 2019-05-25. The design-intent reconstruction — inferred from clinical reasoning alone, before reading the metadata — was correct:

100% osteoarthritis -> the INDICATION

```
(everyone NSAID-eligible)
68% celecoxib + aspirin + ibuprofen    -> the EXPOSURES
the GI differential, all one-shot        -> the ETIOLOGIES
gastrointestinal hemorrhage             -> the OUTCOME
the dataset is literally named GiBleed
```

In this case, the apparent topology has revealed the likely design.

5D · Why this matters in a federated ecosystem

This is where the argument stops being about GiBleed. OHDSI's method is federated (OHDSI, n.d.): an analytic package is written once and executed at many sites against local CDM instances. Patient-level data never moves; only aggregate results return. That architecture has a specific blind spot. If a site's ETL introduces a topology artifact right at the source during extraction such as collapsing recurrent encounters into one record, gating an age range, dropping a mortality field, the network may not see it before data transformation. The site's data passes conformance, the results come back well-formed, and cross-site heterogeneity gets attributed to *population* differences when it may actually be *ETL* differences. That is a confounder introduced by infrastructure, invisible unless subjected to a multi-domain sanity check during DQC.

The implication: **data quality should ideally be established at the source, as opposed to rescued during analysis.** A multidomain sanity check — structure *and* pathophysiology *and* topology — run at each node catches these artifacts before they enter the federation. Run afterwards, all one can do is discover that sites disagree and guess why. This is why the intersection in the Venn is a staffing argument, not a personal one. A CS-trained data-quality engineer will pass this dataset, correctly, by every standard in the toolkit; a clinician without modelling will see the ages are wrong but not that the shape is wrong; a modeller without medicine will see `recs_per_person = 1.00` with no prior that it should be 6. A team needs all three.

6 · Prior work

The broad claim — that synthetic health data lacks clinical realism — is not new. Chen et al. validated Synthea against real-world clinical quality measures in a 1.2-million-person Massachusetts cohort and found limited ability to model heterogeneous post-intervention outcomes (Chen et al., 2019); Wagner and Blacketer (OHDSI Symposium, 2024) compared 1.16 million Synthea-generated California records to real-world data via Achilles and found limited modelling diversity, especially for chronic disease; and the conformance/completeness/plausibility framework GiBleed passes is itself well established (Kahn et al., 2016).

What I believe is narrower and new is fourfold: (1) an audit of **GiBleed specifically** — the

2,694-person default teaching dataset, where most practitioners' mental model of "what OMOP data looks like" is formed, rather than the million-person populations prior work used; (2) **disease-state topology as a data-quality prior** — records-per-person as a signature of transition structure, derived from compartmental-modelling formalism; (3) **module-boundary recovery from output alone** — an undocumented generator module identified via two independent clinical signatures, without source access; and (4) **design-intent reconstruction as a validation method** — recovered from clinical reasoning and then tested against the documented source. Because Synthea's design is documented, this last claim is falsifiable and it held.

7 · Limitations

- **Real-world rates are order-of-magnitude clinical anchors, not the argument.** The point is scale and two figures are, however, exact: the centenarian share ($\sim 0.03\%$ (Schaeffer, 2024)) and adult tetanus-vaccination coverage ($\sim 59\%$ (CDC, 2025)).
 - **n = 2,694.** These findings describe *this* dataset, not Synthea in general; prior work covers Synthea at scale (Chen et al., 2019).
 - **Osteoarthritis and angiodysplasia do occur in midlife** (He et al., 2024; Hudzik et al., 2016). The implausibility is not their presence in the 30s and 40s but the combination of 100% prevalence, hard truncation at 47 years in a population living to 110 years, and complete absence of recurrence.
 - **GI hemorrhage was identified by `condition_concept_id = 192671`** without descendant expansion via `concept_ancestor`.
 - **The module was identified from output only.** I have not read Synthea's source to confirm it corresponds to a discrete generator component. That is the falsifiable part, and I would welcome being checked.
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Figures

Rendered from verified query output; see `figures/`.

- **Figure 1 — 01-topology.png . Records per person, all 31 conditions.** Twenty-three conditions scatter between 1.02 and 6.43; eight sit pinned at the 1.00 floor. The contrast, not the value, is the evidence.
- **Figure 2 — 03-module-boundary.png . Two independent signatures, one boundary, no overlap.** Each bubble is a condition (x = records/person, y = age envelope). Two unrelated rulers draw the same line.
- **Figure 3 — 02-age-envelope.png . Age at diagnosis, all 31 conditions.** Twenty-three span a lifetime to 96–110; the eight occupy a ~ 15 -year band and stop at 47. Viral sinusitis (0–109) is the positive control.
- **Figure 4 — 04-reality-gap.png . Observed versus real-world prevalence (log**

scale). GiBleed prevalences against order-of-magnitude clinical reference values; deaths omitted, as zero has no logarithm.

Reproduce it

```
install.packages(c("Eunomia", "DBI", "RSQLite"))
library(Eunomia); library(DBI)
con <- dbConnect(RSQLite::SQLite(),
getEunomiaConnectionDetails()$server())

# The query that found the module
dbGetQuery(con, "
SELECT c.concept_name,
       COUNT(*) AS n_records,
       COUNT(DISTINCT co.person_id) AS n_people,
       ROUND(COUNT(*) * 1.0 / COUNT(DISTINCT co.person_id), 2) AS
recs_per_person,
       MIN(CAST(strftime('%Y', co.condition_start_date, 'unixepoch') AS
INTEGER) - p.year_of_birth) AS min_age,
       MAX(CAST(strftime('%Y', co.condition_start_date, 'unixepoch') AS
INTEGER) - p.year_of_birth) AS max_age
FROM condition_occurrence co
JOIN concept c ON co.condition_concept_id = c.concept_id
JOIN person p ON co.person_id = p.person_id
GROUP BY c.concept_name
HAVING COUNT(DISTINCT co.person_id) >= 300
ORDER BY recs_per_person DESC;
")
dbDisconnect(con)
```

Portability note. Eunomia's SQLite stores dates as Unix epoch integers rather than the CDM-specified `DATE` type — a small conformance deviation, found by checking a column before trusting arithmetic on it. Hence `strftime('%Y', col, 'unixepoch')`. On PostgreSQL use `EXTRACT(YEAR FROM col)`; on SQL Server, `YEAR(col)`.

Full notebook: `notebooks/gibleed-audit.ipynb` · Standalone SQL: `sql/audit-queries.sql`

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The mathematics tells me what shape a disease should leave in data. The medicine tells me whether the shape aligns to it or not.

In []: